Experiment 02:-

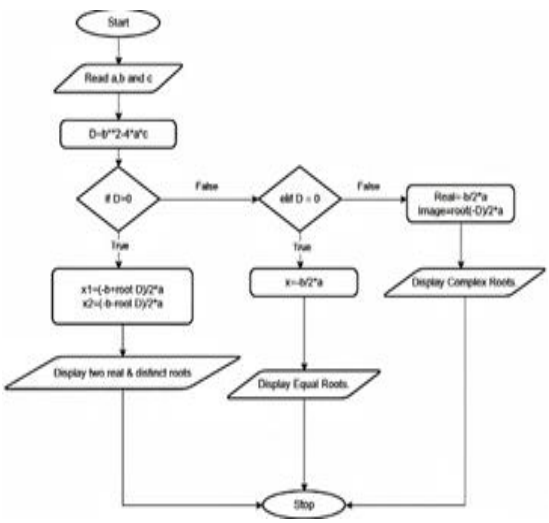
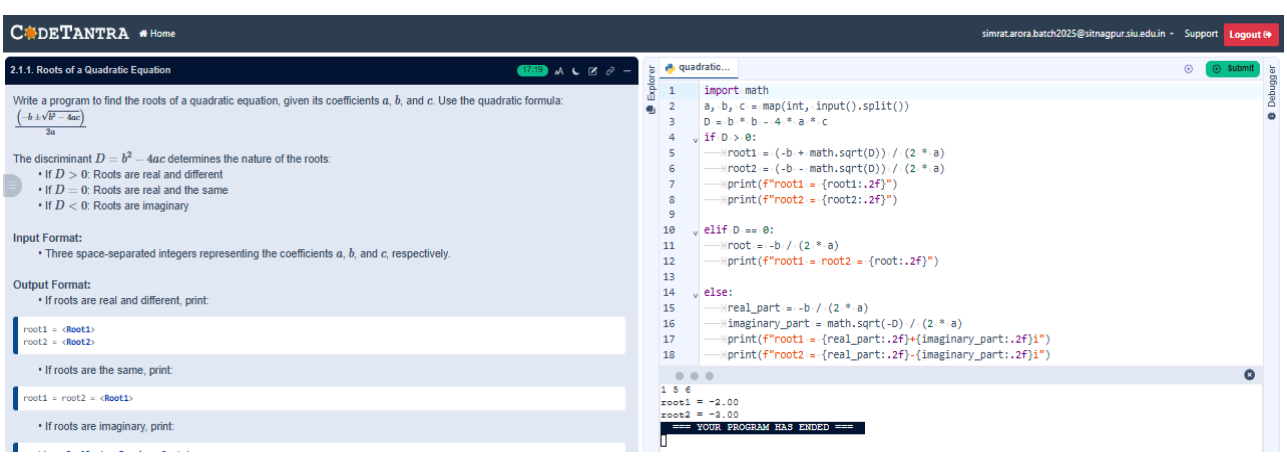
Problem Statement:-

Write a program to find the roots of a quadratic equation, given its coefficients a, b, and c. Use the quadratic formula: (-b+-root(b**2-4ac))/2a.

Algorithm:-

1. Start.
2. Read the coefficients a, b, and c.
3. Calculate the discriminant using:- $D=b$^2−4ac.
4. If D > 0 Compute:- $x1=(-b+\text{root}D)/2a$ and $x2=(-b-\text{root}D)/2a$ Display two real and distinct roots.
5. Else if D = 0 Compute:- $x=-b/2a$ Display equal real roots.
6. Else (D < 0) Compute:- Real part = $-b/2a$ Imaginary part = root(-D)/2a Display complex roots.
7. Stop.

Flowchart:-



Execution:-